PRODUCTION INFRASTRUCTURE V1.2

DOLU LOGISTICS

Complete System Infrastructure & Operations Master Report

This is the **Full Master Overview** of the Dolu Logistics framework (v1.2). This report provides a complete, lengthy, and exhaustive breakdown of the technical, operational, and financial layers of the business. It is designed to be the ultimate reference for management and stakeholders.

The v1.2 Framework

Version 1.2 is our most comprehensive release, moving beyond simple tracking into a fully-integrated **Logistics Command Center**. The system now prioritizes **Real-Time Data Integrity**, **Financial Audit Logs**, and **Field Personnel Safety** through a synchronized web-engine architecture.

1. Technology Stack & Logic Deep-Dive

The system is built for massive scalability in the Port Harcourt logistics market. We use a high-performance cloud stack to ensure zero downtime and sub-second data updates.

Database Layer: Powered by PostgreSQL with Row-Level Security. This ensures that every booking is protected by a hardware-level firewall, preventing unauthorized access to customer data.

Heartbeat Engine: The Rider and Admin portals communicate through a 60-second silent poll. This ensures the "Active Load" is never stale, even if the internet flickers.

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2. Port Harcourt Service Zone & Pricing Utility

Our pricing engine is the core of the business's profitability. It intelligently applies the Highest-Zone Rule to every delivery route.

PostgreSQL v15+ React 18 Framer Motion Lucide Dynamic Icons localStorage Persistence

ZONE RANK	COVERAGE AREAS	STRATEGIC IMPACT
Zone A	Old GRA, Town, D-Line, New GRA	Baseline Urban Logistics
Zone B	Rumuola, Artillery, Woji, Rumuigbo	City Center Distribution
Zone C	Choba, Uniport, Rumuokoro, Eliozu	Educational & Transit Hubs
Zone D	Trans-Amadi, Slaughter, Odili Road	Industrial Corridor Access
Zone E	Eleme, Onne, Eneka, Rukpokwu	Strategic Development Zones
Zone F	Oyigbo, Igbo-Etche, Airport Road, Omagwa	Maximum Distance Premium Tier

Revenue Protection Alert: The system automatically forces the highest zone rate if a pickup and delivery cross multiple zone boundaries.

3. Admin Command Protocols (Master Oversight)

The Admin interface provides the Dispatcher with a centralized hub for every active parcel in the fleet.

- **Dispatch Desk Precision:** When a job is confirmed, the admin must explicitly assign a rider. The system locks the "Rider" field after dispatch to prevent "mid-route rider swapping" errors.
- **Notification Bell:** A visual alarm rings whenever a new "Confirmed" payment is detected. This ensures zero wait-time for the customer.
- **Revenue & Financials:** The Export feature generates a full audit-ready file. Every entry includes the Tracking ID, Rider Name, and Final Price (v1.2 Standard).
- **Fleet Lockdown:** Use the "Archive" tool to instantly disable rider access during off-hours or maintenance periods.

4. Rider Field Operations (Elite Mobile)

The Rider Portal is a high-contrast, tactile interface built for officers moving in fast traffic.

- **Tactile Security Handle:** Riders cannot "mistakenly" click a job as delivered. They must physically **Slide the Lever** across the screen. This confirms a intentional handover.
- **Persistent Session:** The system remembers the rider. If their phone battery dies mid-delivery, they just turn it back on and the dashboard is exactly where they left it.
- **Smart Contact Triggers:** Riders can WhatsApp the Sender for Pickup or the Receiver for Delivery with one tap. The messages are pre-typed for speed and professionalism.

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5. Interactive Parcel Tracking & UX

We provide customers with a **Status-Aware Map**. This build of the system focuses on transparency and visual confirmation.

- **Animated Route Mapping:** The system draws the exact intended path on the customer's map, providing **Visual Proof** of the logistics plan.
- **Pulse Visualization:** The tracking ID pulses to show it is active, and the status bar changes color as the rider swipes their progress.

Standard Operational Workflow

Example Logistics Cycle: Urban to Regional Zone Delivery

1. **Request Phase:** A customer request is initialized from **Old GRA** to **Airport Road**. High-Zone pricing is applied automatically for the outer-zone destination.
2. **Operational Confirmation:** The Administrative team utilizes the Notification Center to verify payment and confirm the booking status.
3. **Fleet Assignment:** The Dispatcher selects an available rider via the Dispatch Desk. The status is advanced to **In Progress**.

4. **Field Notification:** The assigned rider receives an automated notification via the Heartbeat system. They utilize the integrated contact tools for Pickup/Delivery coordination.

5. **Handover Confirmation:** Upon arrival, the rider utilizes the **Tactile Swipe Handle** to confirm physical delivery and handover.

6. **Revenue Finalization:** The digital ledger updates automatically. The customer status is pushed to **Delivered**. 🏁

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6. Technological Roadmap (v1.x Series)

While v1.2 is our current gold standard, our infrastructure is ready for the following future enhancements:

- Digital In-Field Invoicing & SMS Gateways.
- Proof of Delivery (POD) Image Uploads.
- Customer Rating & Performance feedback for Riders.

Dolu Logistics Infrastructure v1.2 Complete